EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession .
DATE:26/03/2025	

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. **Input:**

- The user will enter their **username** and **password** in a **login form** .
- The servlet will validate the credentials and create an **HTTP session** for the user if authentication is successful.

2. **Steps for Implementing Session Tracking with HttpSession:**

1. **Create the Login Form (HTML):**
   - Create an HTML form with two input fields for **username** and **password** .
   - Provide a **submit button** to send the data to the servlet for authentication.
2. **Create the Login Servlet:**
   - Write a Java servlet class to handle **login** requests.
   - The servlet will:
     - Retrieve the **username** and **password** from the form.
     - Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded or from a database).
     - Create a new **HTTP session** for the user if the credentials are valid.

- Set session attributes such as the user's **username** to track the session.

3. **Validate User Credentials:**
   - In the servlet, check if the **username** and **password** match the expected values.
   - If valid, create a session and store the **username** in the session.
   - If invalid, redirect the user back to the login page with an error message.

4. **Create and Track User Session:**
   - Upon successful login, use the HttpSession object to store the session data.
   - Use the session.setAttribute("username", username) method to save the **username** .
   - Optionally, set session timeout using session.setMaxInactiveInterval(time_in_seconds) to manage session expiration.

5. **Redirect to a Protected Page:**
   - After a successful login, the user will be redirected to a **protected page** (e.g., a dashboard or home page).
   - The protected page should check if the user is authenticated by checking the **session attribute** .

6. **Logout and Session Invalidation:**
   - Provide a **logout feature** that invalidates the session using session.invalidate() when the user logs out.
   - After logout, redirect the user back to the login page.

7. **Session Expiration:**
   - If the session expires or the user manually logs out, the session will be invalidated, and the user will be redirected to the login page.

Code:

index.html:

```
<!DOCTYPE html>

<html>

<head>
```

```
    <title>Session Tracking</title>
    <style>
        body {
            background-color: #f3e5f5;
            text-align: center;
            margin-top: 100px;
            font-family: Arial;
        }
        form {
            background: white;
            padding: 20px;
            display: inline-block;
            border-radius: 10px;
        }
        input, button {
            margin: 10px;
            padding: 8px;
        }
        button {
            background: #9c27b0;
            color: white;
            border: none;
            border-radius: 5px;
        }
    </style>
</head>
<body>
```

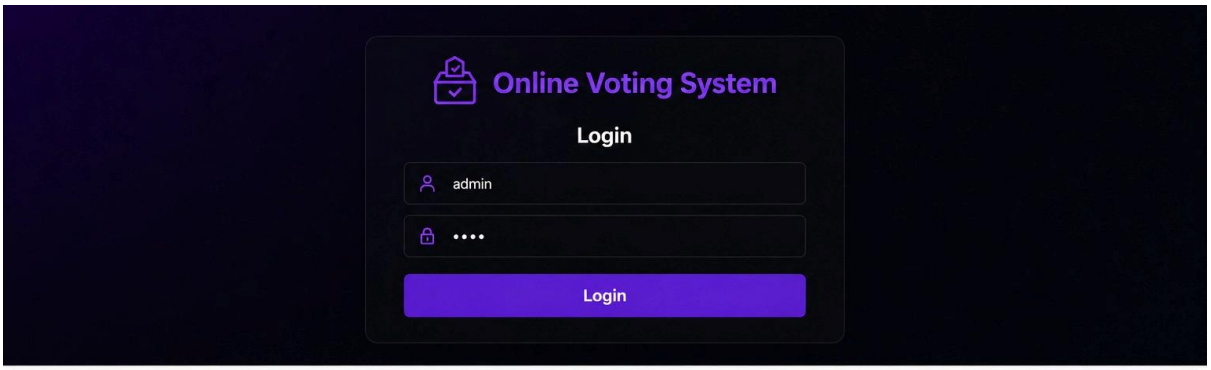
```
<h2>Session Tracking using HttpSession</h2>
<form action="SessionServlet" method="post">
    <input type="text" name="username" placeholder="Enter Name" required><br>
    <button type="submit">Start Session</button>
</form>
</body>
</html>
```

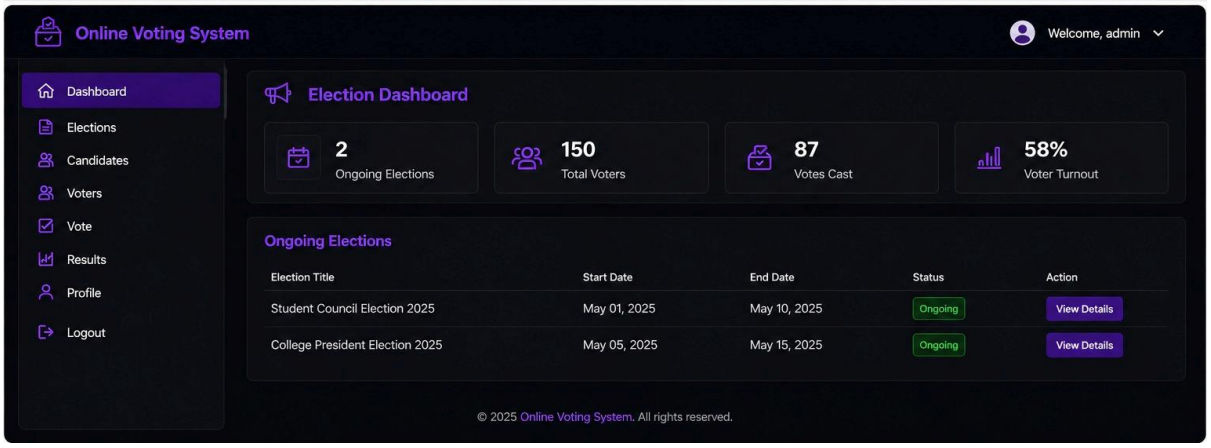
SessionServlet.java:

```
package com.example.session;
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/SessionServlet")
public class SessionServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        String name = request.getParameter("username");
        // Create or get existing session
        HttpSession session = request.getSession();
        // Store data in session
        session.setAttribute("user", name);
```

```
        // Count visits
        Integer count = (Integer) session.getAttribute("count");
        if (count == null) {
            count = 1;
        } else {
            count++;
        }
        session.setAttribute("count", count);
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<html><body style='background:#e1bee7; text-align:center;'>");
        out.println("<h2>Session Tracking Demo</h2>");
        out.println("<p>Welcome: " + name + "</p>");
        out.println("<p>Visit Count: " + count + "</p>");
        out.println("<a href='index.html'>Go Back</a>");
        out.println("</body></html>");
    }
}
```

Output:





RESULT:

A simple login system is created where the user's session is tracked by a servlet to demonstrate session tracking using HttpSession.